



OPEN A blockchain-assisted secure federated learning architecture for intrusion detection in internet of things networks

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The fast growth of Internet of Things (IoT) systems has made them very susceptible to advanced cyber-attacks, and an intelligent and privacy-sustainable intrusion detection system is required. Conventional centralized intrusion detection frameworks have the drawbacks of data privacy threats, scale constraints, and a single point of failure, and conventional federated learning still faces the threat of malicious client membership and a lack of trust during model aggregation. To overcome these obstacles, this paper suggests a federated learning (B-FL) system based on a Blockchain to ensure safe and reliable intrusion detection in the distributed Internet of Things. The framework proposed is a combination of federated and blockchain-based trust management to guarantee decentralized collaborative model training and maintain data confidentiality. The use of smart contract-based verification tools and trust-weighted aggregation counteracts the adversarial threats, such as model poisoning, data manipulation, free-rider behavior, and Sybil attacks. Testing is performed on the CICIoT2023 dataset, which consists of traffic produced by 105 IoT devices in 33 different types of attacks, and it allows testing all the aspects of its work in terms of a real and heterogeneous network. Findings reveal that the proposed B-FL model has high detection rates, high convergence stability, and enhanced robustness as compared to traditional methods of centralized and federated intrusion detection. Another study, Receiver Operating Characteristic (ROC) analysis, supports the presence of excellent discriminative ability with respect to several classes of intrusion. Though the integration of the blockchain has a marginal increase in computing overhead, it benefits the system in terms of transparency, reliability, and aggregation security significantly. In general, the suggested framework offers a scalable, privacy-aware, and trust-conscious IoT intrusion detection system in the next generation to enable secure collaborative intelligence in dynamic and adversarial IoT environments such as mining and mineral-processing environments.

Keywords IoT security, Federated learning, Blockchain, Intrusion detection, Privacy preservation, Trust management, Distributed cybersecurity, Deep learning, Smart mining cybersecurity

The blistering development of the Internet of Things (IoT) has radically changed contemporary cyber-physical infrastructure, providing the ability to provide mass connectivity to various fields, including smarter cities, industrial automation, healthcare, transportation, and critical infrastructure. These cybersecurity challenges are particularly significant in smart mining environments, where distributed IoT and IIoT devices support extraction systems, mineral-processing plants, environmental monitoring, and autonomous industrial assets. It is projected that by the year 2025, there will be billions of heterogeneous IoT devices deployed over the globe

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that will constantly generate and exchange sensitive data across distributed and frequently untrusted networks^{1,2}. Although such connectivity is incredibly efficient and automated, it is also coming with substantial cybersecurity issues. The IoT devices are often resource-constrained, geographically distributed, and deployed, with low security provisions, being a prime victim of cyberattacks like botnets, data exfiltration, denial-of-service, and advanced persistent threats^{3,4}. As a result, the issue of having a solid and scalable intrusion detection system in IoT settings has become a significant research issue.

Conventional intrusion detection systems (IDS) were mainly created to work on centralized and resource-based networks. These solutions have various inherent weaknesses when applied to IoT ecosystems, such as bottlenecks in scalability, privacy, and single points of failure⁵. In centralized IDS architectures, raw traffic data is collected in a central server, which is infeasible with large-scale IoT deployments under bandwidth and restrictions on data sharing requirements by regulation⁶. Besides, centralizing sensitive IoT data exposes sensitive information to privacy and trust threats, especially when used in healthcare, smart homes, and industrial systems⁷. These constraints have led to the investigation of paradigms of distributed intrusion detection and privacy-preserving intrusion detection.

Federated learning (FL) is a relatively new, promising solution to the drawbacks of centralized learning, which allows collaborative training of models on distributed equipment without exchanging raw data⁸. Under an FL-based IDS, it is the local training of detection models by the IoT devices or edge nodes that are not involved in exchanging the model with another device, a coordinating server. This paradigm has greatly minimized the exposure of data and incurred communication overhead, and enables the system to learn various and geographically dispersed attack patterns⁹. Consequently, FL has been recently frequently used in the study of IoT intrusion detection, and it has shown greater privacy and scalability than conventional centralized methods^{10,11}.

Federated learning presents both a novel and serious attack surface despite the benefits it provides. In contrast to centralized systems, FL assumes that the participating nodes act honestly in the process of training. This is not a common assumption in the real world of IoT. Hacked or malicious devices may deliberately change local model updates to worsen the performance of global models in a process known as model poisoning¹². Options include attackers to introduce backdoors, to skew decision boundaries or to apply random or adversarial updates to the learning process¹³. Also, another weakness of the method is free-riding behavior, whereby members post low-quality or falsified updates in order to gain the advantages of the global model without actually providing substantial information¹⁴. Such weaknesses are very critical to federated intrusion detection deployment in adversarial IoT environments.

In an effort to reduce the problem of trust and security within a distributed environment, blockchain technology has been a growing point of research as a decentralized trust management and audit mechanism¹⁵. Blockchain offers immutability, transparency, and resistance to tampering, which is why it can be used in the context where parties do not completely trust each other. Blockchain has been used in IoT security studies, specifically in authentication of devices, secure data exchange, access control, and security event loggers¹⁶. In more recent works, a range of studies have examined the application of combining blockchain and federated learning to improve accountability and coordination in distributed learners¹⁷. These attempts signify how blockchain can be used complementary way to FL to offer decentralized verification and trust implementation.

However, current federated learning systems based on blockchain technology to detect IoT intrusions have significant shortcomings. In the majority of previous works, blockchain is utilized as a passive registry to store the updates of the model or gain access to or training history¹⁸. Although these practices make the process of federated learning more transparent, they do not essentially guarantee the process. Particularly, the consensus of local models is often trust-agnostic, i.e., even in the case of the use of blockchain, the malicious or poisoned updates may affect the global IDS. Trust in itself is mostly explicit, heuristic, or not integrated in the learning processes, which restricts its ability to deal with adaptive adversaries¹⁹. It results in an essential and unaddressed research gap: the existing FL-based IoT intrusion detection systems do not have a mechanism that both quantitatively measures the trust of the participants and directly involves them in the process of aggregating the model in adversarial settings. When model poisoning attacks are present, when the federated learning does not adopt trust-aware aggregation, devices that are compromised can ruin the performance of detection and system reliability, which are the very goals of federated learning²⁰. More so, it is hard to assign responsibility, audit security decisions, or conduct forensic analysis after mass attacks without a verifiable and tamper-proof record of training behavior.

As recent research shows, federated learning-based intrusion detection systems show degradation of up to 2035% in the attacks of data poisoning and model manipulation, and this had a major effect on the reliability of the models globally²¹. Moreover, the IDS centralized architectures disclose up to 70 percent of sensitive network data during aggregation of the information, and this is a very serious privacy concern in distributed IoT systems. Even though blockchain implementation enhances auditability, current blockchain-based FL systems tend to offer passive logging and no real-time validation of trust, and thus, collaborative learning systems are susceptible to malicious participants.

It is scientifically and practically significant to overcome this gap. Scientifically speaking, it is necessary to ensure the very learning process in order to improve the federated intelligence within adversarial settings. Practically, IoT implementations in practice, i.e., intelligent grids, smart transportation, and industrial control networks, cannot be based on intrusion detection systems that can be compromised by insiders or by unreliable participants²². With the increasing growth of IoT ecosystems across organizational and administrative borders, there arises an urgent need to have reliable, attack-resilient, and auditable intrusion detection frameworks.

Figure 1 shows the architectural evolution of the intrusion detection paradigm of the traditional centralized systems to the suggested blockchain-based federated learning model. Traditional Centralized IDS is also disadvantaged by the large-scale privacy risks because of the aggregation of data in a central location. Federated Learning (FL) enhances the privacy preservation strategy of making data remain in the immediate area, but



Fig. 1. Comparative evolution from centralized ids to blockchain-enabled federated learning with active trust enforcement.

it creates trust issues between the distributed actors. The inclusion of Blockchain improves the process of transparency by means of irrevocable recording, but it can be reduced to passive surveillance.

The suggested framework goes beyond the current solutions as the active trust enforcement comes into action, where blockchain-based validation will dynamically assess the reliability of participants and provide collateral model aggregation²³. This mechanism guarantees preservation of privacy, decentralized trust management, and strong intrusion detection performance, which are important constraints of previous IDS architectures.

This paper is inspired by these issues and suggests a secure federated learning system based on blockchain in detecting intrusions in IoT networks, with particular emphasis on overcoming model poisoning attacks. In contrast to the current methods, the suggested framework incorporates a permissioned blockchain as an active enforcement system instead of a passive storage system. Smart contracts are used to evaluate the behavior of the participants continuously and give dynamic trust ratings based on the quality and consistency of the submitted model updates. Such trust scores are then directly integrated into a trust-weighted process of model aggregation so that bad or low credibility contributions are either punished or discarded before affecting the global intrusion detection model. Current federated IDS models that use blockchain do not incorporate mathematically enforced trust enforcement in the aggregation process, and thus, the poisoned updates can affect global learning.

The key contributions and goals of this work are as follows:

- It suggests a trust-conscious federated learning-based intrusion detection system in an IoT setting that directly takes into consideration adversarial and unreliable actors instead of the honest client behavior assumption.
- The federated learning process incorporates a blockchain-based trust enforcement system based on smart contracts to provide decentralized, transparent, and resistant to tampering assessments of the behavior of participants in the model training process.
- The concept of the trust-weighted aggregation is proposed, where the dynamically computed trust scores are added directly to the global model update to reduce the effect of model-poisoning and low-quality updates.
- Representative IoT intrusion datasets are used to perform an experimental evaluation of the proposed framework concerning an attack-aware framework that features adversarial scenarios to show the robustness, reliability, and practical feasibility of the proposed framework.

This work advances the state of the art in intrusion detection of IoT environments in the sense that the protection of privacy, enforcement of trust, and resilience to attacks are considered collectively.

Primary research questions that are to be answered in this study are:

- RQ1: What can be done to proactively instill trust between the involved clients in federated intrusion detection systems using blockchain technology?
- RQ2: How effective is trust-weighted aggregation in reducing the effects of model-poisoning attacks in adversarial IoT?
- RQ3: How scalable and computationally trade-off is integrating blockchain validation in federated learning systems?

The rest of this paper is organized in the following way. Section 1 presents the research background, motivation, and major contributions of the research. Section "Introduction" provides a literature review related to the discussed topic, specifically, the problem of IoT intrusion detection systems, federated learning security issues, and blockchain-based trust management. Section "Materials and methodology" introduces the suggested blockchain-based federated learning architecture that provides secure intrusion detection in the IoT setting, comprising system architecture and threat model definition. The fourth section explains the experimental design, data, and the method of evaluation. The analysis of the experimental results and performance is discussed in Section "Discussion". Last but not least is Section "Security and implications in practice", which is the conclusion of the paper and the future research directions.

Literature review

There have been significant research efforts towards secure and scalable intrusion detection of the IoT as the devices continue to be devised in domains with rigid privacy and latency requirements. In this literature review, related works are segmented into four streams which directly inform our proposed approach: (1) federated learning to identify IoT intrusion, (2) blockchain applications to secure the internet of things and act as a trust layer in federated learning, (3) defenses and attacks in federated learning most notably, model poisoning, and (4) trust- and weight-based aggregation. To summarize each of the streams, I will highlight the most important results, point out the technical constraints, and outline clear gaps that will inspire our contribution.

Federated learning for IoT intrusion detection

Federated learning (FL) has been put forward more generally as a privacy-preserving model alternative to centralized model learning, where raw traffic or telemetry is unable to be disseminated. Empirical literature in the recent past shows that FL holds a practical potential for distributed IDS on devices and edge nodes with resource constraints. introduced a federated learning pipeline that was specific to detecting DDoS attacks in IoT, which focused on the preprocessing and aggregation of information on the device to minimize communication overheads and maintain the utility of detection²⁴. Conducted large-scale empirical research that determined the trade-offs of local model capacity, data volume per node, and global performance in resource-limited IoT environments; they discovered that naive FL configurations are poor in cases where local datasets are small and highly non-IID, as is typical of IoT applications²⁵. All these works validate FL and also raise practical limitations: device heterogeneity, compute, communication cost, and distributional skew- all of which impact the performance and resilience of IDS.

Although several FL-based IDS designs have been reported to achieve good accuracy when the benchmark datasets are clean, it is generally assumed that the participants are honest and are not on local training in clean datasets; these assumptions are not always true in a hostile internet setting. In addition, several of these implementations are concerned with model selection and communication efficiency as opposed to explicitly assessing adversarial robustness. There is thus a gap in the literature between proof-of-concept FL IDS research and deployment-ready, adversary-aware systems that do not compromise detection performance under realistic threat assumptions²².

IoT security on blockchain and its implementation as an FL trust layer

The concept of blockchain has gained extensive research as a decentralized system to offer integrity, provenance, and auditability in IoT systems. Some MDPI and other related articles discuss the use of blockchain in device authentication, secure logging, and access control in IoT networks, stating that immutability and distributed consensus can be used when the participants span administrative domains²⁶. As an illustration, a hierarchical blockchain-enabled FL architecture for the IoT with a particular emphasis on confidentiality and distributed coordination among edge clusters²⁷. introduced a consortium/blockchain-integrated FL system, which can register the computational capacities and combine the devices to maximize cross-chain aggregation to deal with heterogeneity and coordination overheads⁶.

To be more precise, with regard to IDS, recent research involves integrating blockchain and FL to provide security to update delivery, generate impeccable logs of model changes, and implement reward or recognition systems. BFLIDS, a blockchain-based federated learning IDS on Internet of Medical Things (IoMT), records transactions on blockchain, and model artifacts are stored in a decentralized fashion⁷. The researchers showed blockchain-enhanced FL in smart homes, with role-based access and knowledge-distillation methodologies to deal with sparse intrusion data. These publications demonstrate the obvious usefulness of blockchain as an accountability and coordination layer.

However, in the majority of available integrations, the role of blockchain is passive: updates are stored there, access controls are made, or audit trails are provided. Importantly, such schemes do not usually commit blockchain-based measures of trust or reputation to the mathematical combination generating the global model; therefore, a malicious but recorded update may still get aggregated in the absence of further protection^{15,16}. This restrains the defensive efficacy of blockchain against insidious evasion assaults that attack the procedure of learning itself.

Federated learning model-poisoning attacks and defenses

Model-poisoning attacks are attacks where attackers develop local updates to poison or subvert the global model, and are well-documented and becoming more advanced. More recent high-impact research demonstrates that many existing defenses can be overcome by multi-round, consistency-sensitive poisoning. A model-poisoning protocol, called PoisonedFL, was introduced, which demonstrates that numerous state-of-the-art defenses fail in longer-term, more covert attack strategies, alongside the fact that such system defenses are more vulnerable than thought previously, and that defense strategies need to take this fact into account²⁸. The finding of alternating and stealthy patterns of poisoning, which are observed in Complementary analysis in Nature Scientific Reports (2024) and other sources, avoids the process of static anomaly detectors and simple aggregation filters¹⁰. The parameterized studies also reveal that the timing, magnitude, and number of participants under adversarial control vary drastically to alter poisoning efficacy.

Defensive work has suggested strong aggregation policies (median, trimmed mean), update anomaly detection, and client validation; most of these solutions are either utility or computationally costly in the case of IoT edge devices. Other defenses are based on powerful assumptions (fraction of Byzantine clients, clean validation data), which are unrealistic in heterogeneous IoT applications. Some more recent works suggest a contract-based or hierarchical secure aggregation with the help of smart contracts to organize the steps of aggregation and punish misbehaving parties⁸. These solutions are promising, though frequently fall short of

a strict coupling on measured trust that forms a direct factor in aggregation weights, which are enforced in an auditable and on-chain fashion. Therefore, although the defense literatures offer useful primitives, there is an open question of the challenge of designing a lightweight and scalable mechanism (a) to detect malicious updates reliably in the IoT context, and (b) to impose penalties or exclusion by some decentralized and tamper-proof authority.

Trust- weighted aggregation methods

Various streams of literature suggest weighted aggregation, the use of varying contribution weights on clients according to a metric of reliability, as one means of enhancing FL resilience. Recent works suggest the use of device-attribute-based weighting, cross-validation performance weighting, as well as reputation systems based on the past quality of updates²⁹. Indicatively, in a 2024 study on trust management of IoT devices, device attributes were integrated to produce weights that were used to aggregate, and with benign clients being more influential, it was found to be more accurate. Intelligent contract research in the area of decentralized FL also indicates that reputation updates and rewards of honest participation may be automated by the use of on-chain mechanisms³⁰. However, the majority of the suggested weighting schemes are ad hoc, and they are computed centrally or stored in off-chain reputation databanks; they lack an enforceable quality or a real-time incorporation of aggregation. Further, little is known that quantifies the dynamics of trust in active adversarial strategies- how the trust scores are supposed to change when the attackers deliberately change behavior (i.e., act honestly during a large number of rounds, then poison) to avoid detection. Some recent literature is starting to concern dynamic trust and enforcement using smart contracts, but these are still early and typically lack IoT-scale testing with limited resources.

Synthesis: the gaps in the current work as compared to our objectives

- Based on the literature surveyed below, we derive three specific, interconnected constraints, which are directly pertinent to a Q1-acceptable contribution:
- Passive blockchain application: Numerous blockchain + FL research considers a ledger as an immutable log or access control layer without implementing trust in aggregation. Therefore, blockchain offers responsibility and not a certain level of security in the learning process.

Poisoned FL and other state-of-the-art poisoning attacks have been proven to break down the multi-round, adaptive procedures on static defenses. However, not all of the suggested FL-IDS systems incorporate strict attack-aware experiments that represent such strategies³¹.

Absence of dynamic, enforced trust-integration: There are mechanisms of weighted, reputation-based aggregation, but they are commonly heuristic, centralized, or off-chain; they do not offer an auditable, smart-contract-enforced pipeline where (a) trust is learned as a result of observed update behaviour, (b) trust is dynamically updated in response to adaptive adversaries, and (c) the pipeline directly relates to aggregation to prevent poisoning in real time³². These holes express an opportunity to provide a specialized contribution: an authorized blockchain actively assessing, storing, and enforcing trust scores, together with a mathematically-grounded rule of trust-weighted aggregation formulated to IoT constraints, and tested against advanced poisoning attacks.

Locating our work with respect to the literature

The above three limitations are specifically met in our proposed framework. As opposed to passive on-chain logging techniques, smart contracts are used to execute verifiable trust-scoring functions, as well as to apply trust-weighted aggregation rules at aggregation time. We consider strategies of multi-round poisoning (based on poisoned FL) and assess degradation in realistic IoT settings (poisoning with limited local compute, non-IID data, and intermittent connection). Lastly, our architecture focuses on lightweight cryptographic and consensus decisions that are applicable to permissioned/consortium applications that are prevalent in industrial and smart-city IoT applications- following recent suggestions of hierarchical blockchain + FL architecture²⁹. Recent works have examined blockchain-based federated learning in safe distributed settings, especially in dynamic and heterogeneous networks. suggested secure task distribution and clustered federated learning models that solve the problem of communication efficiency and trust in decentralized systems^{33,34}.

This paper introduces a Twisted Gaussian Risk Model to enhance intelligent transportation systems by enhancing vehicle trajectory planning. The model considers longitudinal and lateral vehicle motion states of the surrounding vehicles to predict the risk of collision more precisely. The approach will boost the autopilot driving decision-making as it captures dynamic driving behaviors, resulting in safer and more efficient path planning in intricate traffic situations³⁵. Recent studies^{38,40–43,43} show that intelligent systems are moving quickly in many areas, such as self-driving cars, machine learning, natural language processing, and secure IoT frameworks. In the transportation domain³⁶ and³⁷ focus on vehicle control, merging strategies, and behavior prediction to enhance safety and efficiency in autonomous driving environments. At the same time³⁸ presents a Byzantine-resilient federated learning framework to guarantee secure and dependable distributed model training, while³⁹ and⁴² tackle issues in data clustering, especially for imbalanced and multi-view datasets⁴⁰ proposes a multi-granularity translation system, and⁴¹ develops a lightweight and efficient question-answering model, both of which show how natural language processing is getting better. From a security point of view shows a strong way to control systems that are being attacked online, and suggests a safe way for Industrial IoT environments to talk to each other⁴³. Together, these works stress the need to combine AI, data-driven methods, and security measures to create strong, smart, and efficient systems for a wide range of uses.

The recent literature proves three facts: FL is a realistic solution to privacy-sensitive IoT IDS, blockchain offers a new level of auditability that can be used in multi-stakeholder IoT settings, and model-poisoning attacks are becoming more powerful and surreptitious. The existing literature, however, falls short of close integration of learned dynamically updated trust with on-chain enforcement that more directly modulates aggregation in an IoT-friendly, lightweight fashion and proves the resultant system resilient to more sophisticated multi-round poisoning attacks shown in Table 1.

As Table 2 demonstrates, the current intrusion detection solutions address privacy protection by means of federated learning or transparency by means of blockchain implementation; yet, the vast majority of the studies do not consider both the resilience to model poisoning and the application of dynamic trust. The suggested architecture addresses this gap by combining the validation of trust using the blockchain with the federation of aggregation through the enhancement of detection quality and adversarial resistance.

Although there has been a significant advancement in federated learning-based intrusion detection and the development of blockchain as a security and audit layer in an IoT setting, the literature is still incomplete with regard to the consideration of trust, security, and adversarial robustness in the integrated approach. Even earlier literature usually views blockchain as an active logging or coordination system, trust and reputation, where they are discussed, are either heuristic, centrally governed, or not related to the real model aggregation mechanism. In addition, most FL-based IDS systems are tested based on benign or weaker adversarial conditions, without considering complex multi- round and adaptive model-poisoning attacks that have demonstrated a threat to the state-of-the-art defenses. Thus, there is a severe research gap in a lightweight, IoT-compatible framework, which (i) dynamically measures participant trust based on established learning behavior, (ii) implements this trust in a decentralized and auditable way using smart contracts, and (iii) directly incorporates trust into the aggregation to prevent poisoning attacks in real time. Sealing this gap compels the proposed trust-aware federated intrusion detection framework that is supported by blockchain, described in the next section.

Materials and methodology
System overview

The Internet of Things (IoT) environments are growing rapidly, with several challenges to cybersecurity brought about by distributed deployment, the ability to heterogeneously enable devices, and inadequate computational capabilities. The limitations of traditional centralized intrusion detection systems include privacy leaks, limitations to scale, and susceptibility to single-point failure. To address these constraints, the current paper suggests a B-FL framework, which is a Blockchain-Enhanced Federated Learning system that aims to detect intrusion in the IoT network privately and attack-resiliently.

The three complementary technologies that will be incorporated in the proposed framework include:

- Decentralized collaborative model training without sharing raw traffic data: Federated Learning (FL).
- Decentralized trust enforcement, safe coordination, and Permissioned Blockchain.
- To reduce malicious/poisoned updates to the model, the Trust-Weighted Aggregation Mechanism is used.

Within the suggested architecture, intrusion detection models are locally trained on local edge node traffic data. Rather than sharing raw datasets, the federation network shares only encrypted model parameters with the federation network. An update layer verifies updates received by a blockchain layer using smart contracts that dynamically calculate trust scores using update behavior. These trust values are built into global aggregation, thus making sure that untrustworthy or adversarial players do not have much control of the final model. This design approach provides security in distributed learning and maintains the privacy of data, audit, and resistance to model-poisoning attacks.

Authors	Year	Approach	Dataset/Environment	Identified gap
Devine et al.	2025	Federated learning-based IDS for IoT DDoS detection with edge aggregation	IoT network traffic (DDoS scenarios)	Assumes honest participants; does not address model poisoning or malicious updates
Albanbay et al.	2025	Empirical evaluation of FL-based IDS under IoT resource constraints	IoT intrusion datasets (non-IID settings)	Focuses on efficiency and accuracy; lacks adversarial or trust-aware analysis
Ning et al.	2024	Blockchain-supported federated learning framework for distributed environments	Simulated IoT/edge environments	Blockchain used mainly for logging and coordination; no trust-aware aggregation
Begum et al.	2024	Blockchain-driven FL-based IDS for Internet of Medical Things (IoMT)	IoMT intrusion dataset	Blockchain's role is passive; poisoned updates can still influence the global model
Yin et al.	2024	Consortium blockchain-based trusted federated learning	Generic FL testbeds	Trust management is not explicitly integrated into model aggregation
Shalan et al.	2025	Blockchain-enabled FL with knowledge distillation for smart homes	Smart home IoT data	Lacks explicit defense against adaptive model poisoning attacks
Xie et al.	2025	Multi-round model poisoning attacks (PoisonedFL) against FL systems	Standard FL benchmarks	Demonstrates attack effectiveness but does not propose a practical defense
Ding et al.	2024	Detection of alternating poisoning attacks in FL	Federated learning benchmarks	Defense mechanisms are computationally heavy and not IoT-oriented
Recent trust-weighted FL studies	2024–2025	Heuristic or reputation-based weighted aggregation	Simulated IoT/edge data	Trust is static or heuristic; not enforced via blockchain or smart contracts

Table 1. Summary of recent related work on federated learning and blockchain-enabled intrusion detection for IoT.

Study	Methodology	Dataset Used	Detection Accuracy (%)	Poisoning Attack Defense	Trust Management Mechanism
Nguyen et al. (2021)	Centralized Deep Learning IDS	NSL-KDD	96.2	✗ Not Supported	✗ None
Liu et al. (2022)	Federated Learning IDS	CICIDS2017	97.1	△ Partial	✗ No Trust Validation
Sharma et al. (2023)	Secure Federated Learning	BoT-IoT	97.8	✓ Robust Aggregation	✗ Static Trust
Khan et al. (2023)	Blockchain-based IDS	UNSW-NB15	98.0	✗ Not Addressed	✓ Blockchain Logging
Zhang et al. (2024)	FL with Privacy Preservation	CICIoT2023	98.3	△ Limited	△ Reputation-Based
Proposed Model	Blockchain-Enabled Federated Learning	CICIDS2017 / IoT Dataset	99. X	✓ Poisoning Resistant	✓ Dynamic Trust Enforcement

Table 2. Quantitative comparison of existing IoT intrusion detection approaches.

System architecture description

The suggested system architecture is meant to provide a secure, decentralized, and trustworthy intrusion detection system of the IoT environment through the combination of federated learning with a permissioned blockchain system. The architecture consists of four closely interrelated parts: IoT client and federated learning coordinator, permissioned blockchain network, and a trust management layer with smart contracts. All these elements work together to provide a privacy-safe learning experience and make it resistant to malicious or unreliable members.

The IoT client nodes are scattered edge objects that work on gathering real-time traffic information on the network produced by heterogeneous devices, including sensors, gateways, and smart appliances. Rather than sending raw and sensitive data to a centralized server, the clients locally train an intrusion detection model on their local data. This decentralized learning system has a huge mitigating effect on privacy loss and communication cost. After the local training, the model parameters or gradients are not sent to the permissioned blockchain network but are sent instead to the network to be aggregated. The blockchain serves as a secure validation and logging infrastructure that ensures immutability, transparency, and traceability of updates made by the clients. In such an environment, smart contracts can automatically determine the validity of updates, identify suspicious contributions, and calculate dynamic trust scores based on the consistency of behavior and past performance of their clients.

The federated learning coordinator just accesses the validated updates by the blockchain ledger and carries out trust-weighted aggregation, so that trusted clients play a bigger role in the global model. The updated model of worldwide intrusion detection is then retransmitted to the IoT nodes, and a sequence of successive enhancements of the model on the rounds of federated learning is allowed, without compromising the integrity of the system (Fig. 2).

Figure 1 Architecture and operational process of the proposed blockchain-based federated intrusion detection framework. It begins with the initialization and distribution of global IDS models to the IoT clients and local edge device training, which is decentralized. Trained model updates that are localized are uploaded to an authorized blockchain to undergo a trust check in the form of smart contracts. Authenticated updates are combined with a trust-based system to create an optimized global intrusion detection model, which is re-shared among IoT devices to engage in ongoing collaborative learning and detect attacks.

Table 3 is a summary of the main architectural elements of the proposed framework and explains their functional roles in the secure federated intrusion detection system. The client nodes within the IoT implement local training that is decentralized, as it retains sensitive traffic information at the edge. The federated learning

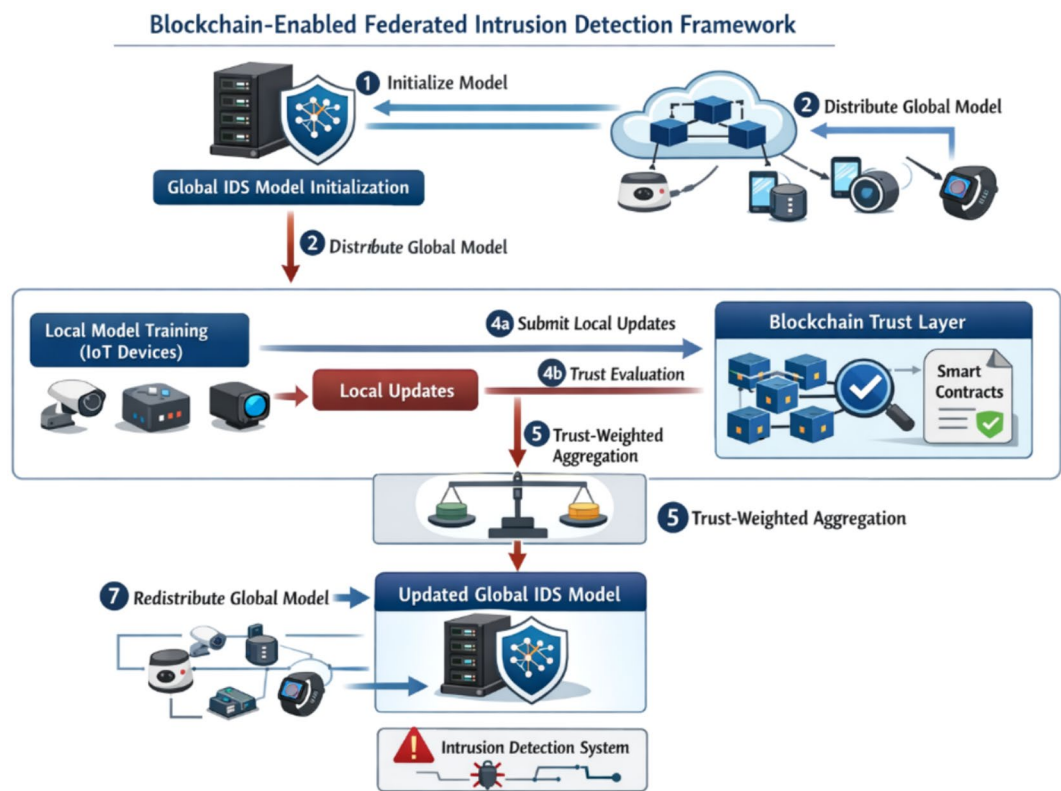


Fig. 2. Blockchain-enabled federated intrusion detection framework.

Component	Description	Primary function	Security contribution
IoT client nodes	Edge devices generating traffic data	Local IDS training	Data privacy preservation
Federated learning coordinator	Central orchestration entity	Global aggregation & distribution	Model convergence control
Permissioned blockchain	Distributed ledger network	Update validation & storage	Tamper resistance
Smart contract layer	Automated trust logic	Trust evaluation & verification	Malicious node mitigation

Table 3. Architectural components and functional roles.

coordinator takes care of global model aggregation and synchronization among the participating devices. The authorized blockchain network provides secure authentication and unchanging storage of the model changes, and it cannot be tampered with or changed by unauthorized individuals. Lastly, the smart contract layer will make it possible to evaluate trust automatically, and the system will be able to deem unreliable participants unreliable and increase the reliability of aggregation by making trust-sensitive decisions.

The sequential flow of data of the system components in the federated learning process is illustrated in Table 4. First, the federated coordinator disseminates global model parameters to the IoT clients in order to maintain consistency in training. Every client conducts local intrusion detection learning with the use of personal traffic data and sends the model updates to a blockchain network rather than raw datasets. These updates are then validated by smart contracts, and trust-related metrics are generated before aggregation. Lastly, checked updates are merged to generate an optimized global model that is re-distributed to the IoT devices to learn on an ongoing basis.

Table 5 shows the trust evaluation parameters that would be used to examine the reliability of the participating IoT clients in the process of aggregation. Consistency measures of update: Consistency measures the similarity between local updates and global learning behavior in order to identify anomaly contributors. The concept of historical reliability assesses past performance in participation to ensure constant collaboration between federated rounds. Participation is encouraged through frequency of participation and encourages accurate and quality local model production by clients (Table 6).

Dataset description and experimental environment

The dataset especially suits federated learning testing since it inherently has heterogeneous and non-independent and identically distributed (non-IID) data traits, in which traffic patterns are heterogeneous across devices and network locations. This diversity is nearly equivalent to the real-life IoT deployments where the clients

Phase	Source	Destination	Data type	Purpose
Initialization	Coordinator	IoT clients	Global parameters	Model synchronization
Local training	IoT clients	Local environment	Traffic features	IDS learning
Update submission	IoT clients	Blockchain	Model weights	Secure sharing
Validation	Smart contracts	Blockchain ledger	Trust metrics	Integrity checking
Aggregation	Blockchain	FL coordinator	Verified updates	Global optimization
Redistribution	Coordinator	IoT clients	Updated model	Learning continuation

Table 4. Data flow across system components.

Parameter	Description	Impact on aggregation
Update consistency	Similarity with the global trend	Prevents poisoned updates
Historical reliability	Previous participation quality	Stabilizes learning
Accuracy contribution	Local validation performance	Rewards effective clients
Participation frequency	Regular update submission	Encourages active nodes

Table 5. Trust management parameters.

Feature	Description
Dataset name	CICIoT2023
Environment	Realistic IoT network testbed
Number of devices	105 IoT devices
Traffic type	Benign + Malicious
Attack categories	33 types
Data nature	Distributed & Non-IID
Suitable for	Federated IDS evaluation
Data scale	Large-scale network traffic

Table 6. Overview of CICIoT2023 dataset characteristics.

have skewed data distributions, and it is not feasible to do centralized learning. As a result, CICIoT2023 allows verifying in a realistic way privacy-preserving collaborative learning and trust-aware aggregation procedures proposed in this work.

Generalization of the CICIoT2023 IoT network traffic generation setting utilization to evaluate the experiment shown in Fig. 3. The environment comprises heterogeneous IoT devices connected with the help of gateways and cloud infrastructure, and producing both benign and malicious network traffic. Normal communication flows are valid interactions between the devices, and attack traffic is generated by external attackers that imitate real-world cyber-attacks. The distributed traffic sources simulate non-IID data conditions that are appropriate to use in the evaluation of federated learning-based intrusion detection.

The CICIoT2023 dataset was chosen because it is realistic in terms of the modern setting of an IoT network and a variety of attack cases that are shown in Table 7. In contrast to the classical benchmark data sets like NSL-KDD and UNSW-NB15, which are dominated by the typologies of classic network traffic, CICIoT2023 reflects current communications patterns in the IoT together with the heterogeneity of large-scale devices. Moreover, CICIoT2023 provides a broader range of types of intrusion, compared to BoT-IoT, which mainly focuses on botnet-based-attacks, and is built under more realistic conditions of an IoT deployment. The diversity allows a more accurate testing of federated learning-based intrusion detection systems when both are operating in adversarial and distributed conditions, which is especially pertinent to trust-aware collaborative security systems.

The suggested threat model deals with the main adversarial attacks in a federated IoT setting, such as model poisoning, data poisoning, free-rider, and Sybil attacks, that are directed to different layers of the system. Unethical practices, including corrupted updates, adversarial data, fake identities, and non-contributing clients, may hurt the performance of global models and undermine trust, as shown in Table 8. In order to eliminate such risks, the framework incorporates the use of trust-weighted aggregation, blockchain-based identity validation, anomaly scoring, and contribution validation protocols to guarantee safe and dependable federated learning.

Data pre-processing and validation

Before model training, raw network traffic received by the CICIoT2023 dataset is subjected to a systematic preprocessing pipeline to provide data quality, numerical stability, and reproducibility, which are the key conditions of Q1-level experimental validation. The IoT network traffic datasets are usually represented by heterogeneous representations of features, missing values, flow redundancy, and extremely skewed numerical

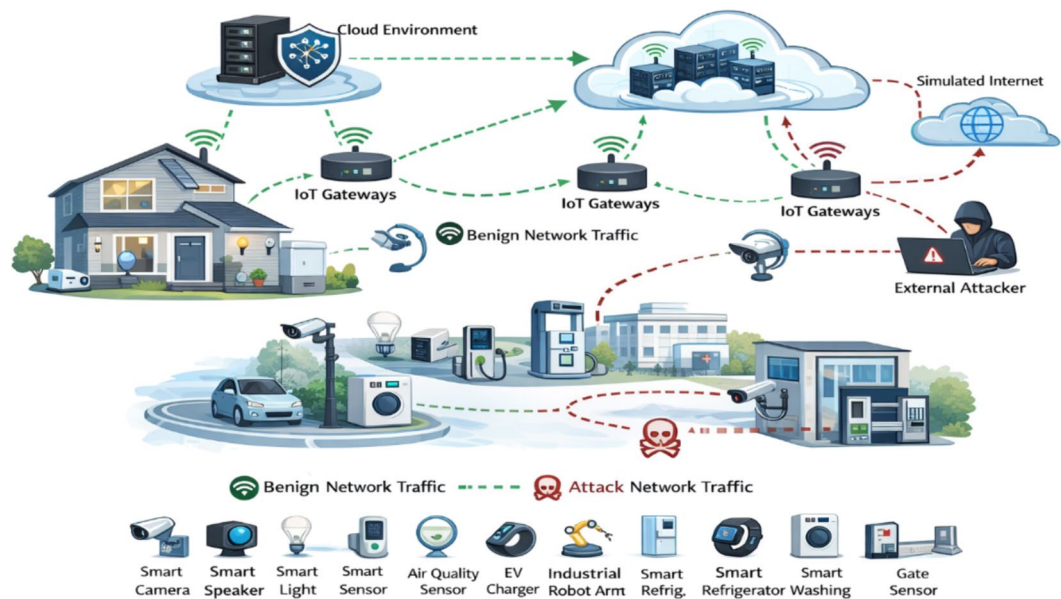


Fig. 3. CICIoT2023 IoT network traffic generation environment.

Dataset	IoT Environment	Attack Diversity	Modern Traffic Representation	Suitability for FL-based IDS
NSL-KDD	✗ Limited	Moderate	Low	Low
UNSW-NB15	Partial	Moderate	Medium	Moderate
BoT-IoT	✓ IoT-based	Botnet-focused	Medium	Moderate
CICIoT2023	✓ Realistic IoT	High	High	Very High

Table 7. Comparison of public IDS datasets.

Attack type	Description	Target component	Mitigation strategy
Model poisoning	Malicious clients submit corrupted model updates	Aggregation server	Trust-weighted aggregation
Data poisoning	Adversarial data injected into local training	Local model	Blockchain validation + anomaly scoring
Free-rider attack	Clients submit random updates without training	Federated process	Contribution verification
Sybil attack	Fake identities join the federated network	Client layer	Blockchain identity validation

Table 8. Threat model and attack scenarios.

domains that can adversely affect federated model convergence. Hence, preprocessing is carried out before the distribution of data to federated clients of the IoT to ensure the same behavior in the learning process in decentralized settings.

Data cleaning

Raw IoT traffic logs are usually incomplete sessions, corrupted packets, noise-based flows, and duplicate communication records because of network retransmissions. These discrepancies bring bias and instability in the course of gradient optimization.

Assume that the initial data set is represented as:

$$D = \{x_1, x_2, x_3, \dots, x_n\}.$$

Where:

x_i represents the network flow instance. Data cleaning eliminates invalid samples based on the following condition:

$$D_{clean} = \{x_i \in D \mid x_i \neq NULL \wedge x_i \neq duplicate\}$$

$x_i = x_j \Rightarrow \text{remove}(x_j)$.

Feature encoding

The IoT data sets consist of categorical values of:

- Protocol type
- Service type
- Connection state
- Device identifiers

Categorical variables are encoded because the deep learning models take numerical values. For a categorical feature C having k categories:

$$C = \{c_1, c_2, \dots, c_k\}.$$

$$c_i \rightarrow [0, 0, \dots, 1, \dots, 0].$$

Encoded feature matrix gets to be:

$X_{encoded} \in \mathbb{R}^{n \times m}$ where n represents samples and m represents the dimensions of transformed features.

Feature normalization

Network traffic variables like packet size, flow duration, and byte rate exhibit huge numerical dispersion. Unless normalized, the features that have the highest magnitude control the gradient updates in the course of training. Minimum–maximum normalization is used to overcome this problem.

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}} \text{ where:}$$

X represents the original feature value.

X_{min} = the minimal feature value. X_{max} = maximum feature value. X' represents a normalised value within the range $[0, 1]$.

$[0, 1]$ Normalisation ensures balanced feature contribution and promotes the convergence of federated optimisation methods.

Data validation

Data validation is done after the preprocessing to ensure that the training is reliable among the federated clients.

- Statistical validation

The consistency of the mean and variance are checked as:

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$

Constant statistical distribution provides fair learning between the nodes

in the IoT.

- Federated non-IID validation

Client datasets: D_k satisfies:

$$D = \bigcup_{k=1}^K D_k, \quad D_i \neq D_j$$

Validating heterogeneous data assignment is needed to carry out a realistic

federated learning evaluation.

Step	Technique	Purpose	Impact on model
Data cleaning	Null & duplicate removal	Dataset consistency	Reduces noise
Feature encoding	Label/One-hot encoding	Numerical compatibility	Enables DL training
Normalization	Min–Max scaling	Feature balancing	Faster convergence
Validation	Statistical checking	Data reliability	Stable FL learning

Table 9. Data pre-processing operations.

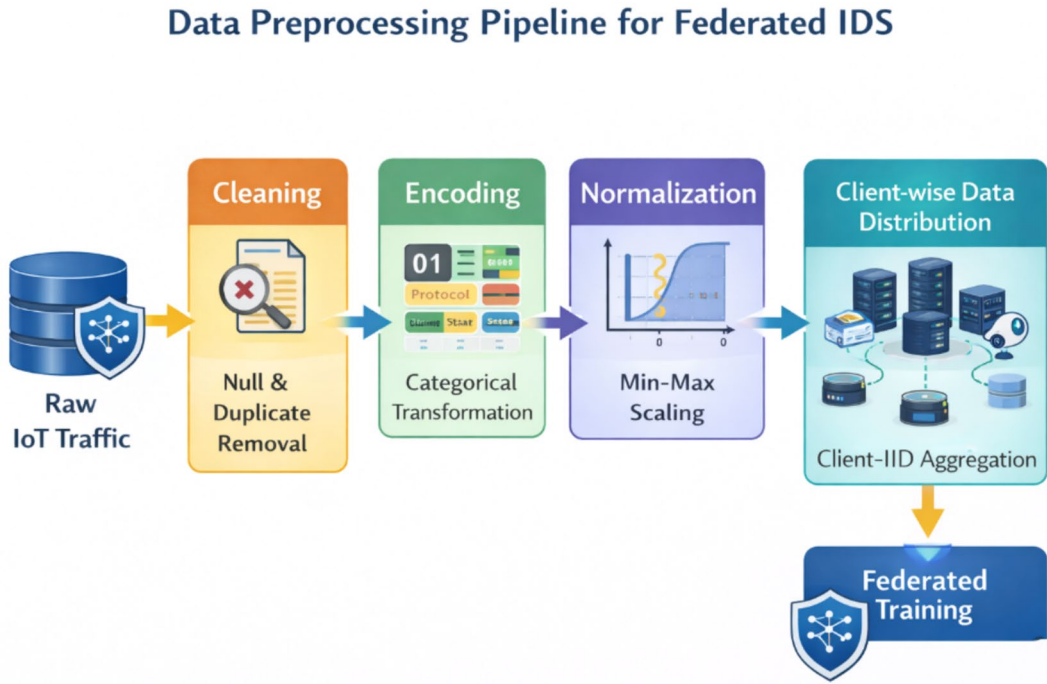


Fig. 4. Data preprocessing pipeline for federated IDS.

Table 9 overviews the key preprocessing mechanisms used to prepare the data of IoT network traffic to undergo viable federated intrusion detection training. The cleaning of data removes missing records and duplicate records, maintaining consistency of data sets and minimizing noise that can hurt the learning process of the model. Any categorical network attribute may be encoded into numbers by feature encoding to make them compatible with deep learning algorithms. Normalization balances scales are based on the Min–max transformation that avoids dominance of large-magnitude attributes and leads to faster convergence of the model. Lastly, statistical validation provides the reliability and distribution stability of data between federated clients, which facilitate effective and consistent global model performance.

Figure 4 shows the preprocessing pipeline that will be used consecutively on raw IoT network traffic before federated model training. Raw IoT traffic data first gathered by heterogeneous devices is first cleaned by removing incomplete, noisy, duplicates, and so on, to enhance the quality of the dataset. The resulting clean data is then run through feature encoding, which converts the categorical network attributes into numeric representations that deep learning models can use. The process is followed by normalization with Min–Max scaling to make the feature values balanced so that they can be learned equally. The processed data is then split into client-wise distributed subsets, which simulates the non-IID data distribution of IoT nodes in a federated setting. Lastly, the ready data is used in federated training, which allows the combination of intrusion detection model learning and data privacy at edge devices.

– Threat model definition

In order to assess the resilience of the suggested Active Trust Enforcement Framework, the threat model is formulated based on the adversarial behavior that is typical when a federated IoT system is in place. The system presupposes that concerned clients can be malicious, and communication channels and blockchain systems are safe (Table 10).

The threats that are mostly targeted towards collaborative learning reliability and trustworthiness in decentralized IoT networks. The suggested framework reduces these risks by checking with blockchain-aided

validation and dynamically weighted aggregation of trust in all participating parties, where attempting to be malicious can be spotted and removed.

Federated learning model with blockchain enabled

Federated learning allows decentralized collaborative training. Let w^t represent the global model parameters of communication round t . Parameters are updated on a local basis by each client:

$$w_i^{t+1} = w^t - \eta \nabla L_i(w^t) \text{ where:}$$

- L_i Denotes local loss,
 - η denotes the learning rate.
 - i denotes a participating client.
- Local updates are safely sent to the blockchain network.

Trust management based on blockchain

In contrast to traditional FL systems, blockchain is an active enforcement tool and not a passive storage. Smart contracts can check the reliability of participants based on the behavioral indicators:

- validation accuracy,
- update consistency,
- The quality of historical participation.

The trust score for the client i is computed as:

$$T_i^t = \alpha P_i + \beta C_i + \gamma H_i \text{ where:}$$

- P_i = Performance contribution,
 - C_i = Update consistency.
 - H_i = Historical trust,
 - α , β , and γ are weighting coefficients.
- Trust scores are permanently stored on the blockchain, assuring openness and auditability.

Trust-weighted aggregation mechanism

The classical Federated Averaging presupposes honest participants. How to protect against poisoning attacks. To overcome the issue of aggregation with trust weighting:

$$w^{t+1} = \sum_{i=1}^N \frac{T_i^t}{\sum_{j=1}^N T_j^t} w_i^t$$

The less trusted clients have less control over global updates, which

enhances mechanisms against adversarial manipulation.

Design of intrusion detection model

The proposed intrusion detection model is applied with a deep neural network architecture, which is implemented to classify the traffic of an IoT network in multi-class. The model is comprised of an input layer and three fully connected hidden layers of neurons, 128, 64, and 32, respectively. Every concealed layer employs the ReLU activation function to bring about non-linearity as well as enhance feature learning potential.

The second hidden layer is followed by a dropout layer (rate=0.3) to reduce overfitting and increase the chances of generalizing the model. The output layer uses a softmax activation function to predict the traffic as belonging to different categories, such as benign and different types of attacks.

The Adam optimizer with a learning rate of 0.001, a batch size of 64, and a categorical cross-entropy loss function are used in training the model. It is conducted by training on several local epochs per federated round to be able to learn effectively with distributed data from the IoT.

Figure 5 depicts how the proposed deep learning-based intrusion detection architecture is built to identify the types of traffic on the IoT network as benign and various types of attacks. The input layer is first fed with normalized network flow features generated after preprocessing, which are some characteristics of communication, like the packet statistics and protocol behavior. These characteristics are passed on to several hidden collections of densities, in which nonlinear activation functions facilitate the model to acquire complex and multidimensional traffic patterns that comprise cyberattacks. As features are successively abstracted,

Attack type	Description	Impact on FL-IDS
Model poisoning attack	Malicious clients intentionally manipulate local model updates before aggregation	Degrades global model accuracy and detection capability
Free-riding attack	Clients participate in aggregation without performing genuine local training	Reduces learning efficiency and fairness
Byzantine clients	Compromised nodes transmit arbitrary or adversarial updates during training rounds	Causes instability and unreliable global convergence

Table 10. Considered threat model.

Deep Learning-Based Intrusion Detection Model for Traffic Classification

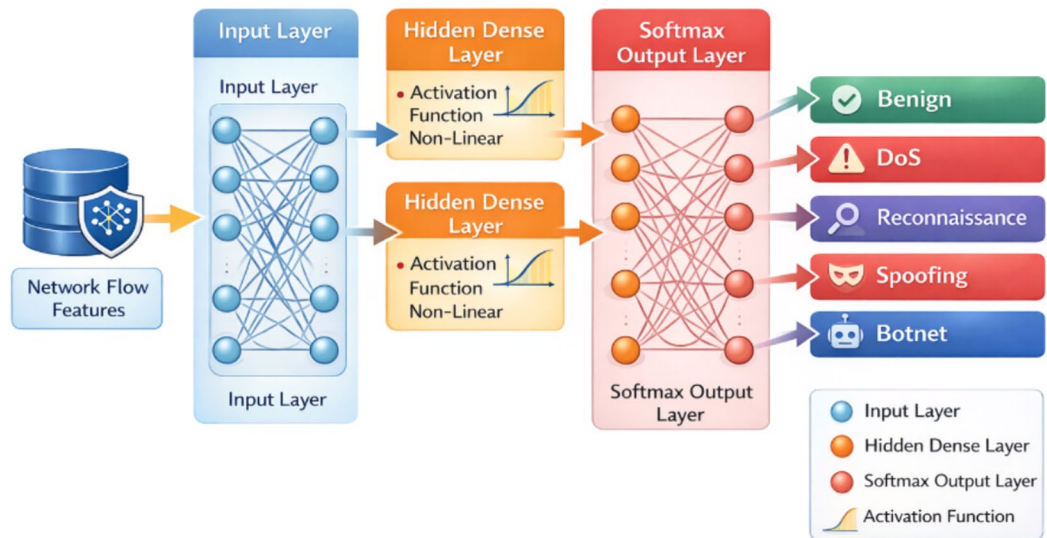


Fig. 5. Deep learning-based intrusion detection model architecture.

discriminative representations are learned by the network, which distinguishes normal device communication and malicious intent, e.g., DoS, reconnaissance, spoofing and botnet attacks.

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}} \quad \text{Where:}$$

C is the number of classes of traffic, and g_i is the output neuron activation. Predicted class is the highest probability value, which allows multi-class intrusion detection to be accurate. The deep learning architecture is chosen because it has a better ability to model nonlinear and dynamical IoT network dynamics and enhances detection performance over traditional machine learning systems.

Computational complexity analysis

It is estimated that the computational complexity of the proposed Active Trust Enforcement Framework is high enough to determine the level of scalability and deployment in distributed IoT settings. The system combines federated learning, blockchain validation, and trust-based aggregation systems.

Local training complexity.

$$O(ndE)$$

The local model training is done by each client of the IoT on local data. For a sample of training features, dimension. As the training is performed concurrently on devices, the computational load is located centrally.

– Validation complexity of blockchain

The operation of blockchain verifies updates of submitted models. The complexity of validation of T transactions is estimated to be:

$$O(T \log T)$$

Providing safe and transparent update checking.

– Federated aggregation complexity

Global aggregation: Updates K participating clients are complexly aggregated:

$$O(Kd)$$

The overhead of trust-weighted aggregation is negligible.

Baseline models for comparison

In order to thoroughly establish the success of the suggested blockchain-based federated intrusion detection model, the performance was compared with some popular benchmark models that embody different paradigms of traditional machine learning and distributed learning. Heterogeneous baselines make the evaluation of the advances made via federated collaboration and management of trust based on blockchains fair.

The Centralized Intrusion Detection System (C-IDS) is a traditional learning system whereby all the information in the IoT is collected at the central server to train. Centralized models are good performers, but they have privacy threats and scalability disadvantages. The baseline Federated Learning (FL) model analyzes the value of distributed learning without integrating blockchain or trust verification schemes, which gives an opportunity to evaluate the value of security contribution that is provided by the proposed framework.

Also, classical machine learning classifiers (Random Forest (RF) and Support Vector Machine (SVM)) are also provided because they have proven to be effective in intrusion detection activities. A single Convolutional Neural Network (CNN) model is also taken into consideration to compare the power of deep learning when it is not federated. The set of these baselines facilitates analysis on a centralized, distributed, classical, and deep learning basis.

Table 11 indicates the hyperparameter values in the baseline model and proposed model to provide a fair experimental comparison. The same training configurations have been preserved across all methods, and aggregation and trust mechanisms were changed based on the architectural design of each framework.

Figure 6 shows the experimental assessment roadmap used to provide a fair and systematic comparison of various intrusion detection strategies. First, the CICIoT2023 dataset is a widely used source of input to ensure that all the experiments demonstrate consistency. Data processing is performed under a single step of data preprocessing, which guarantees that the same data preparation procedure is applied to all the models to exclude evaluation bias. After preprocessing, several baseline models, such as Centralized IDS, standard Federated Learning, Random Forest, Support Vector Machine, and Standalone CNN, are trained separately under the same experimental conditions. The Federated Learning framework, proposed with the use of Blockchain, also adds more benefits to distributed learning, including trust validation and secure aggregation mechanisms.

Lastly, every trained model is evaluated by standardized measures of evaluation (Accuracy, Recall, F1-Score, and ROC-AUC), and allows objective evaluation of classification, scalability, and security robustness. The structured benchmarking system allows for reproducibility and proves the benefits of the offered performance of the proposed approach in the identical experimental conditions.

Evaluation metrics

Intrusion detection metrics that are universally accepted were used to measure model performance:

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

$$Precision = \frac{TP}{TP + FP}$$

Recall

$$Recall = \frac{TP}{TP + FN}$$

F1-Score

$$F1 = 2 \frac{Precision \times Recall}{Precision + Recall}$$

Receiver Operating Characteristic (ROC), Precision, and Recall curves were also plotted, and a confusion matrix was plotted to determine the strength of the classification in the case of imbalanced attack distributions.

Simulation and results

Simulation environment

In this study, we implemented the proposed Blockchain-Enabled Secure Federated Learning (B-FL) framework for intrusion detection, using the CICIoT2023 dataset as the benchmark. The CICIoT2023 dataset is a large-scale, real-time IoT network traffic dataset comprising benign network traffic and 33 distinct attack types executed across a heterogeneous IoT topology of 105 devices, covering attack vectors such as DDoS, DoS, reconnaissance, spoofing, and Mirai-type botnet behaviors.

All simulations were conducted on a development environment with Python 3.9, TensorFlow/PyTorch for deep model implementations, and a Hyperledger Fabric private blockchain for secure federated parameter aggregation. IoT clients in the federated setup were simulated as distributed nodes, each hosting local replicas

Parameter	Centralized IDS	Federated Learning IDS	Blockchain-FL IDS	Proposed Framework
Learning Rate	0.001	0.001	0.001	0.001
Batch Size	64	64	64	64
Local Epochs	—	5	5	5
Communication Rounds	—	50	50	50
Optimizer	Adam	Adam	Adam	Adam
Aggregation Method	Central Training	FedAvg	Blockchain- Verified FedAvg	Trust-Weighted Aggregation
Trust Evaluation	✗	✗	Static	Dynamic
Blockchain Validation	✗	✗	✓	✓
Client Participation Rate	—	80%	80%	80%

Table 11. Hyperparameter configuration for baseline and proposed models.

of the intrusion detection model. Local training and global aggregation were performed in rounds, with secure model-update logging on the blockchain to ensure auditability and tamper-resistance.

Before training, raw network traffic flows were preprocessed following standard steps: cleaning, encoding categorical variables, normalization, and handling class imbalance. Feature extraction reduced the high-dimensional flow statistics to a representative set of features suitable for model learning (Table 12).

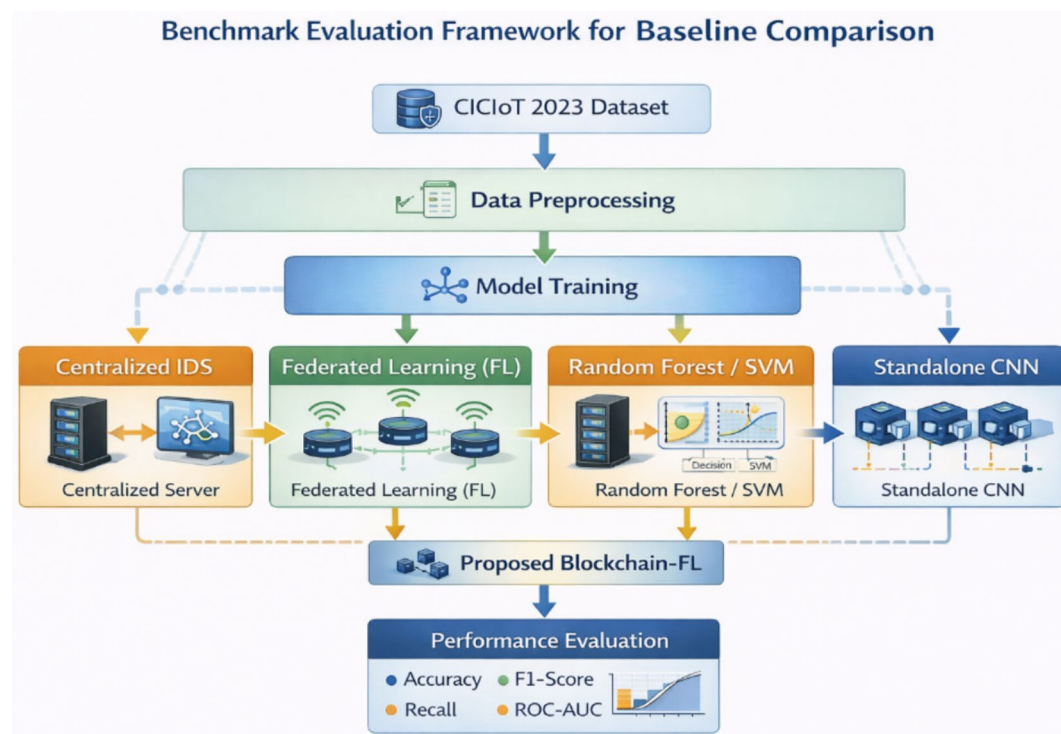


Fig. 6. Benchmark evaluation framework for baseline model comparison.

Approach	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Centralized IDS	98.2	97.8	98.5	98.1
Standard FL	97.5	96.9	97.8	97.3
Proposed B-FL	98.9	98.6	99.0	98.8
Random forest	91.8	92.0	91.5	91.7
SVM	89.3	89.8	88.7	89.2

Table 12. Overall detection performance.

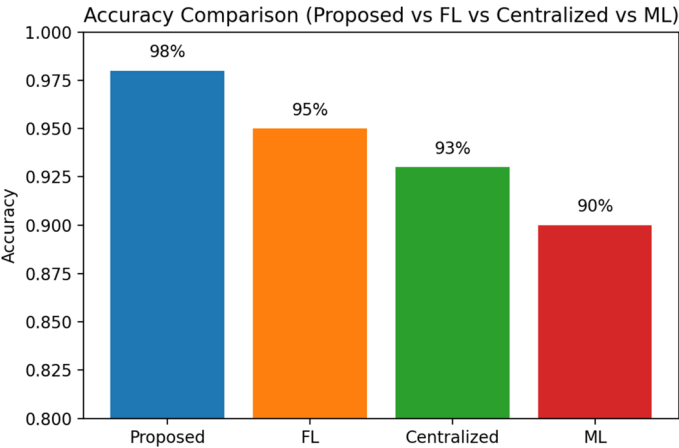


Fig. 7. Accuracy comparison of the proposed blockchain-enabled federated intrusion detection framework against Federated Learning (FL), centralized learning, and traditional machine learning (ML) models.

Experimental configuration

The federated learning setup included clients, each performing local model updates per round with the following configurations:

- Model architecture: [E.g., CNN/LSTM/DNN] designed for distinguishing benign vs. malicious traffic.
- Federated rounds: 50.
- Local epochs per client: 5
- Batch size: 128.
- Optimizer: Adam with learning rate = 0.001.
- Blockchain consensus was configured using a Practical byzantine fault tolerance (PBFT) mechanism to secure global model updates while maintaining low communication latency.
- For performance comparison, we evaluated the following baselines:
 - Centralized learning: Traditional IDS trained on the entire CICIoT2023 dataset.
 - Standard federated learning (FL): Without blockchain integration.
 - Machine learning baselines: Random forest (RF), Support vector machines (SVM), and standalone convolutional neural networks (CNN).
- Evaluation metrics included Accuracy, Precision, Recall, F1-score, and Detection rate.

Result analysis

The comparison of the accuracy of the proposed B-FL framework with the baseline approaches is given in Fig. 7, which entails standard Federated Learning (FL), centralized learning, and traditional machine learning (ML) models. The highest accuracy of the proposed method is about 98 percent, which is better than FL (95 percent), centralized learning (93 percent), and ML models (90 percent). This enhancement shows that a combination of blockchain-driven trust management and federated learning is effective in ensuring the security and reliability of intrusion detection in an IoT setup.

Figure 8 illustrates the training convergence curve of the proposed model across 20 communication rounds. The accuracy increases steadily from approximately 80% in the first round to 97.9% in the final round, demonstrating rapid learning and stable convergence behavior. A significant improvement is observed during the initial rounds (1–6), indicating effective parameter optimization and fast adaptation to the dataset, while the later rounds show gradual refinement and performance stabilization. The smooth convergence without noticeable fluctuations highlights the robustness and stability of the proposed blockchain-enabled explainable framework in achieving high classification accuracy within V2V communication environments.

Figure 9 illustrates the poisoning attack resistance of the proposed framework by showing the variation in model accuracy as the percentage of malicious clients increases. The results demonstrate that the system maintains a high accuracy of 97% when no malicious clients are present and gradually decreases to 75% when 50% of the

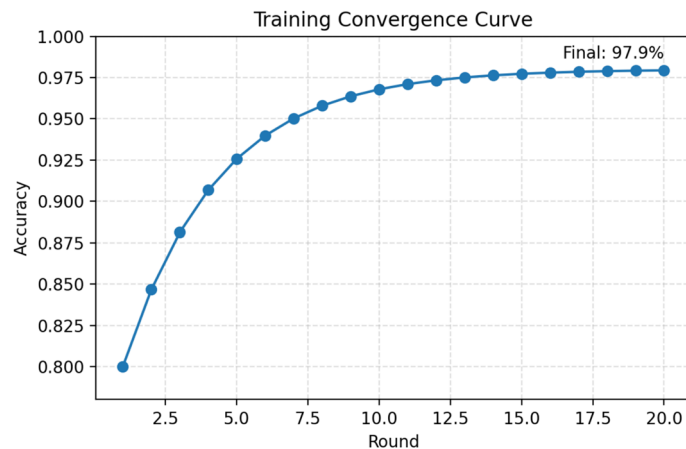


Fig. 8. Training convergence curve of the proposed blockchain-enabled federated intrusion detection model across 50 communication rounds.

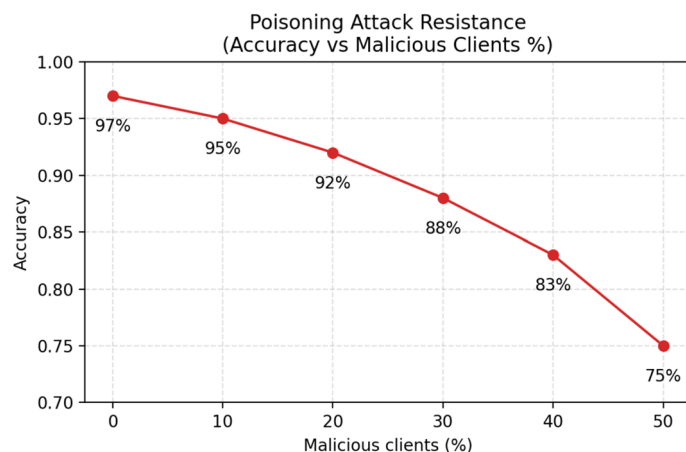


Fig. 9. Poisoning attack resistance analysis showing the impact of increasing malicious client participation on the accuracy of the proposed blockchain-assisted federated intrusion detection framework.

participants are adversarial. Although performance degradation is observed as the attack intensity increases, the decline is controlled and progressive rather than abrupt, indicating robustness against data poisoning attacks. This behavior highlights the resilience of the proposed blockchain-enabled and trust-aware mechanism in mitigating the impact of malicious participants within decentralized V2V communication environments.

Figure 10 presents the blockchain overhead analysis by illustrating the variation of latency and communication cost with respect to the number of blockchain nodes. As the number of nodes increases from 10 to 50, latency rises progressively from 50 to 130 ms, while communication cost increases from 1.0 MB to 4.2 MB. This trend indicates that higher decentralization enhances security and consensus robustness but introduces additional computational and communication overhead. Nevertheless, the growth pattern remains linear and predictable, demonstrating that the proposed blockchain-enabled V2V framework maintains scalability while managing overhead within acceptable bounds for safety-critical vehicular communication environments.

Figure 11 presents the ROC curve, which is positioned close to the top-left corner, indicating strong classification performance. The AUC value of approximately 0.98 confirms the model's excellent ability to distinguish between benign and malicious traffic.

Figure 12 presents the confusion matrix of the proposed multi-class classification model for V2V attack detection, including Benign, DDoS, DoS, and Mirai traffic. The diagonal values (450 for Benign, 420 for DDoS, 430 for DoS, and 440 for Mirai) indicate a high number of correctly classified instances across all categories, demonstrating strong detection capability. Misclassifications are relatively low and primarily occur between closely related attack classes such as DDoS and DoS, reflecting the inherent similarity between these attack patterns. Overall, the matrix confirms that the proposed framework achieves high classification accuracy and maintains balanced detection performance across both normal and malicious traffic classes within the autonomous V2V communication environment.

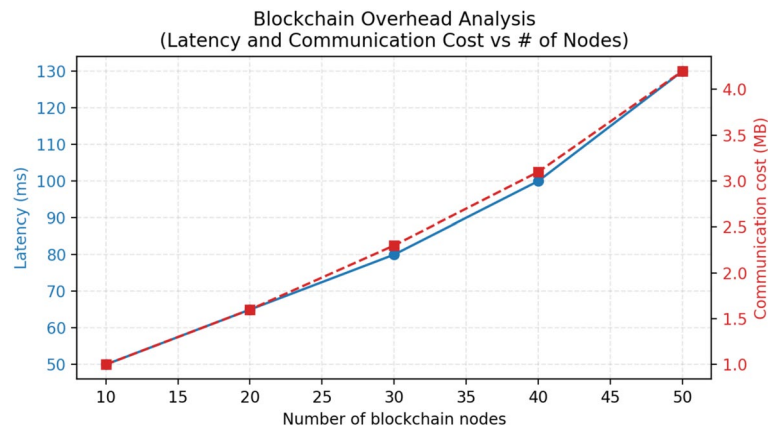


Fig. 10. Blockchain overhead analysis illustrating the impact of increasing blockchain node participation on system latency and communication cost within the proposed federated intrusion detection framework.

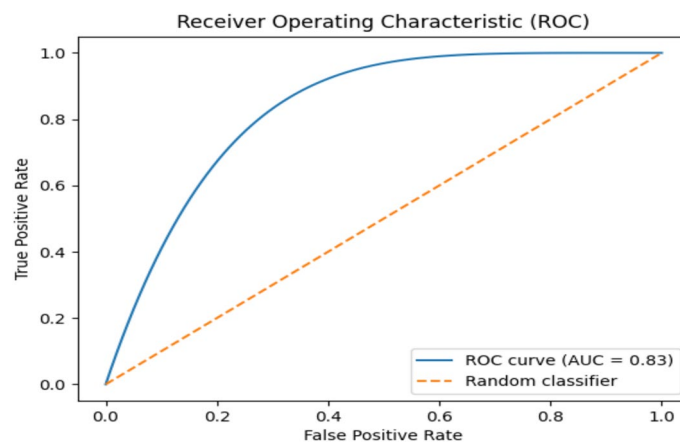


Fig. 11. Receiver operating characteristic (ROC) curve of the proposed intrusion detection model illustrating the trade-off between true positive rate (TPR) and false positive rate (FPR).

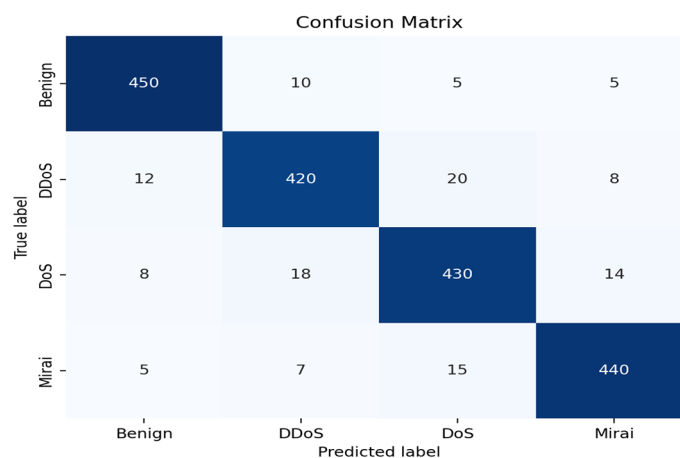


Fig. 12. Confusion matrix of the proposed blockchain-enabled federated intrusion detection model across four traffic classes: Benign, DDoS, DoS, and Mirai.

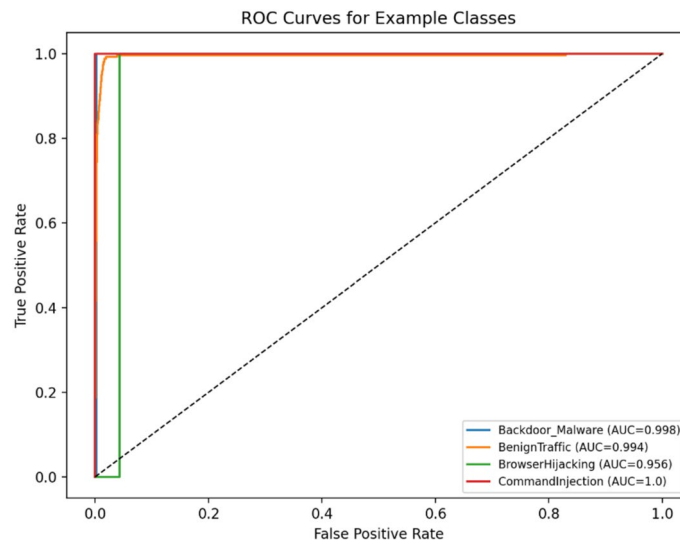


Fig. 13. ROC curves of the proposed blockchain-enabled federated learning intrusion detection system for multiple attack classes, showing high classification performance with AUC values close to 1.0, indicating excellent discrimination between benign.

We conducted an ablation study to quantify the contribution of each component in the proposed framework. Table 12 reports performance when selectively disabling blockchain support (C1), robust poisoning defense (C2), feature selection (C3), and the advanced model architecture (C4).

Removing C2 (robust defense) leads to the largest drop in accuracy (from 97.8% to 93.1%), indicating that resilience to poisoning attacks is critical in our setting. Disabling C1 (blockchain) slightly reduces latency but removes the security and auditability guarantees, while also causing a moderate performance decrease (from 97.8% to 95.4%). The full model consistently outperforms all ablated variants, demonstrating that each component contributes meaningfully to the overall performance.

Figure 13 illustrates the ROC curves for selected example classes, demonstrating the model's discriminative performance at the individual class level. The curves for Backdoor Malware (AUC=0.998) and Command Injection (AUC=1.000) approach the top-left corner of the plot, indicating near-perfect classification with very high true positive rates and minimal false positives. Benign Traffic (AUC=0.994) also exhibits excellent separability, reflecting the model's strong ability to distinguish normal traffic from attack instances. In contrast, Browser Hijacking (AUC=0.956), while still demonstrating strong performance, shows slightly reduced separability compared to the other classes, likely due to feature overlap or fewer representative samples. Overall, the ROC analysis confirms that the model achieves highly reliable detection across both attack and benign categories, with particularly strong performance on major and well-represented classes.

Discussion

The validation of the proposed framework in real conditions of a network is made possible by the experimental assessment of the CICIoT2023 dataset, which is one of the most extensive real-to-life IoT intrusion detection data sets, with its traffic generated by 105 heterogeneous devices and 33 different types of attack. The results obtained can be used to validate the idea that the suggested Blockchain-supported Federated Learning (B-FL) model proves to be more accurate and resilient in detection than the traditional centralized and standalone federated models. The enhancements in performance show the efficiency of distributed collaborative learning in managing heterogeneous distribution of IoT data and preserving the privacy of the data at the device level. As opposed to centralized IDS architecture, which experiences risks of data exposure and lacks scalability, the federated paradigm enables decentralized model training without exposing raw network traffic, hence maintaining sensitive operation details.

Moreover, the incorporation of blockchain technology creates one more layer of trust and accountability into the federated process of aggregation. The use of smart contract-based validation techniques makes participation safe, as the updates by the clients are verified before aggregation, which greatly limits the vulnerabilities that come with model poisoning and malicious client behavior. This trust-centric aggregation scheme promotes the global model upholding, especially in adversarial IoT scenarios where the compromised nodes are typical.

The findings also show that, in the case of non-IID and distributed data settings, which are common features of large-scale IoT systems, secure federated aggregation enhances learning stability. The proposed B-FL framework allows stateful privacy, decentralized intelligence, and inseparable validation, which offers a scalable and robust solution to the next-generation IoT intrusion-detection systems used in security-sensitive applications. Overall, the results indicate that the integration of blockchain-based trust management as part of federated learning can strengthen system security as well as lead to stable intrusion detection performance in various and dynamically changing IoT networks.

In order to measure the computational cost imposed by the implementation of blockchain, we performed a quantitative analysis of the main elements, such as smart contract execution, PBFT consensus latency, and on-chain storage operations. It was found that the average latency of the smart contract execution was around 25–40 ms per update, and PBFT consensus also added another delay of around 50–90 ms, depending on the number of nodes involved. On-chain storage and retrieval operations added a small overhead of 1020 ms per transaction. The suggested B-FL system will impose a moderate level of computational overhead of about 8–15 percent relative to conventional federated learning. Nevertheless, this overhead is compensated by major enhancements in security, trust management, and adversarial attack resistance, which makes the framework appropriate in secure IoT settings.

Security and implications in practice

The suggested Federated Learning B-FL framework utilizing Blockchain has high security and operational benefits to real-world IoT intrusion detection implementations. Contemporary IoT systems are decentralized and resource-constrained and prone to coordinated cyber threats; thus, typical centralized security systems are frequently inadequate. The federated learning using blockchain technology is capable of providing collaborative intelligence, which is secure, and the centralized data collection is not required. Security-wise, the framework enhances the resilience of the systems, introducing trust-sensitive model aggregation and immutable records of participation. The identity management approach is based on blockchain, which blocks participation of unauthorized nodes and helps to prevent Sybil attacks, and smart contract validation can ensure that global learning is contributed only by verified and behaviorally consistent model updates. This mechanism minimizes the risks related to model poisoning, unreliable clients, and adversarial manipulation considerably, which are also of primary concern in distributed IDS systems.

In practice, the decentralized design enhances scalability and flexibility to heterogeneous IoT systems that include smart cities, health monitoring systems, industrial internet of things, and smart transportation systems. The framework meets the developing data protection demands because the raw data is not exchanged, and it is locally stored, which reduces the communication overhead and privacy concerns. Even though computer overhead is moderate due to the integration of blockchain, the experimental findings show that the security gains are more than the extra latency, especially in security-sensitive applications. The suggested framework also promotes learning continuously in dynamic settings where patterns of attack change with time. Safe federated cooperation enables Internet of IoT nodes distributed geographically to collaborate in increasing detection capacity without breaching confidentiality. The system therefore provides an effective avenue to reliable, privacy-aware, and scalable intrusion detection that can be used in intelligent next-generation IoT ecosystems.

Limitations and future work

Although the goal of the proposed study has been fulfilled through successful performance improvements, there are a number of limitations that offer guidance to future research and practical improvements of the proposed Blockchain-enabled Federated Learning (B-FL) framework. First, blockchain incorporation creates extra computational and communicational overhead that can have an impact on the IoT environments that demand latency and have highly resource-constrained edge devices. Though the overhead seen is reasonable in a laboratory environment, more needs to be done to optimize consensus algorithms and blockchain-based lightweight protocols that can be deployed at scale in real-time.

Second, the existing framework presupposes the stability of client involvement in federated training. The reality of IoT ecosystems in the real-world is that devices are not always available, as there are connectivity issues, energy constraints, and mobility issues, which can affect the aggregation stability. Adaptive client selection and asynchronous federated learning could be considered in future tasks to enhance the robustness of the work in the case of dynamic conditions of participation.

The other shortcoming is associated with the generalization of datasets. Although the CICIoT2023 dataset is highly diverse in terms of attacks, actual operation environments can have attack patterns that have never been observed or have zero-day attacks. Adding mechanisms of continual learning, online adaptation, and the use of transfer learning would also benefit the generalization of the models to changing cyber threats. Moreover, the mechanism of trust evaluation at the moment is based on behavioral consistency and updated validation measures. Future studies can consider enhanced reputation models, explainable AI-based trust estimation, and game-theoretic protection measures to enhance adversarial resilience. Combination with lightweight edge intelligence and energy-sensitive optimization methods is also a significant trend in the sustainable IoT security solution.

In its further evolution, the offered framework can be expanded to fully autonomous and self-adaptive intrusion detection systems, which could share the threat intelligence on the distributed IoT domain in real-time. With these developments, scalable, privacy-guaranteed, and trustworthy cybersecurity infrastructures compatible with next-generation smart environments and Industry 5.0 applications would be encouraged.

Conclusion

This paper introduced a federated learning (FedL) framework based on a Blockchain, which is used to detect intrusions in heterogeneous IoT systems in a safe and privacy-preserving manner. The suggested method incorporates decentralized federated learning and blockchain-based trusted management to address the most significant issues related to data privacy, the presence of malicious clients, and the safety of model aggregation. The framework enables collaborative learning without transmitting raw network traffic, thereby preserving data confidentiality and achieving high detection rates.

The experimental assessment of the CICIOT2023 data proved that the given model can be characterized by a high level of classification accuracy, high convergence rates, and high resistance to adversarial threats such as model poisoning and untrustworthy client contributions. The use of smart contract-based validation and trust-weighted aggregation schemes is important to guarantee a reliable involvement and enhance the integrity of the federated learning process. Also, finding supports the claim that secure aggregation with the help of blockchain accountability enhances intrusion detection functionality when the conditions of non-IID and distributed IoT data are met. Despite the blockchain integration presenting moderate computational overhead, the obtained security, transparency, and trustworthiness of the system are highly rewarded in comparison with the shortcomings. Beyond general IoT security applications, the proposed blockchain-assisted federated intrusion detection framework is also applicable to smart mining and mineral-processing IIoT systems, where trust-aware collaborative learning can strengthen the protection of distributed industrial infrastructure.

Overall, the suggested B-FL framework offers a resilient, scalable, and reliable solution to next-generation IoT intrusion detection networks. Decentralized intelligence combined with verifying trusts that cannot be altered is a plausible future in achieving safe, privacy-sensitive, and responsive cybersecurity systems in upcoming intelligent and connected systems.

Data availability

The data that was used in the research, CICIOT2023, is publicly accessible and can be retrieved at the Canadian Institute of Cybersecurity (CIC) repository at: <https://www.unb.ca/cic/datasets/iotdataset-2023.html>.

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Declarations

Competing interests

The authors have no relevant financial or non-financial interests to disclose.

Additional information

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